

Response to Reviewer' Comments for Manuscript [25000741]

We sincerely thank the reviewer for the valuable and constructive comments. We have carefully revised the manuscript and addressed each point raised. Below, we provide detailed responses to the reviewer's comments, with corresponding modifications made in the revised paper highlighted accordingly. We hope the revisions have sufficiently addressed the concerns and improved the quality of the manuscript.

Reviewer C

[Major comment#1]

Section 4.1 says "Experiments 1 and 2 deal with static scheduling problem, while Experiment 3 deals with dynamic scheduling problem.", but the authors added the new Experiment 3 (Fig.19) to the manuscript. Therefore, Section 4.1 should explain the four experiments.

Answer:

We sincerely thank the reviewer for pointing out this important oversight. We apologize for the omission in the previous revision. As the reviewer correctly noted, Experiment 3 was newly added as an ablation study comparing IA-NSGA-II with IA-NSGA-II (Fix) and IA-NSGA-II (Non-VNS). However, its description was mistakenly not included in Section 4.1. In the revised manuscript, we have carefully revised Section 4.1 to clearly explain all four experiments, including the objective and setting of Experiment 3. We appreciate the reviewer's careful reading and helpful comment, which has helped us improve the clarity and completeness of the manuscript.

Specific modifications in the manuscript:

From revised paper (Section 4.1):

Through numerical experiments, we verify the usability of the IA-NSGA-II proposed in this paper. Experiments 1 and 2 deal with static scheduling problem, and Experiment 3 is an ablation study comparing IA-NSGA-II with IA-NSGA-II (Fix) and IA-NSGA-II (Non-VNS), while Experiment 4 deals with dynamic scheduling problem. In Experiment 1, we select a test benchmark instance from the reference Wu et al. [22] and compare the results of IA-NSGA-II with the original scheduling method "INSGA-II" proposed in Wu et al. [22]. In Experiment 2, we select a test benchmark instance from Wu et al. [27] in the context of FAJSP-AGV and compare the results of IA-NSGA-II with those of the scheduling method "IGA" proposed in Wu et al. [27] and traditional NSGA-II.

In Experiments 1 and 2, the solution was obtained independently for each instance to ensure reliability and consistency; Experiment 1 was run 10 times and Experiment 2 was run 20 times. The experiment 3 was conducted using the same test instances and parameter settings as in Experiment 2. In Experiment 4, three of the 20 scheduling results from Experiment 2 were randomly selected and the rescheduling method described in Section 3.5 was used. The common parameters for the algorithm in all experiments were set as follows: initial population size 100, number of genetic generations 100. The algorithm is implemented using the programming language MATLAB.

[Major comment#2]

Fig.19 shows the search process of the IA-NSGA-II (Proposal Method; red line), the IA-

NSGA-II (Fix; green line), and the IA-NSGA-II (Non-VNS; blue line) and the IA-NSGA-II is superior to the others. Although there is no difference between three methods in first generation, the initial objective values (left end of the line in Fig.19) differ greatly from each other, i.e., the blue line is 90, the red line is 85, and the green line is 80. Therefore, I suspect that initial populations (a random seed) are not fixed. Please add the sentence to Section 4.4, for instance, "The same initial population is used to examine the differences in the search process of each method." Moreover, if the authors do not fix initial populations in Experiment 3, the experiment should be retried to use the same initial solution for a fair comparison. And, other experiments should be reviewed as well.

Answer:

We sincerely thank the reviewer for pointing out the inconsistency in the initial makespan values presented in Fig. 19. As correctly noted, the three curves in the original figure started from different initial makespan values, which may raise concerns regarding the fairness of comparison.

Upon careful review, we found that this discrepancy was caused by the application of the variable neighborhood search (VNS) operator immediately after initialization in the IA-NSGA-II and Fix variants. This local search process improved the initial solution quality before generation 0, resulting in lower initial makespan values. In contrast, the Non-VNS variant did not include this local search step, leading to the observed difference at the start of the curves.

We agree with the reviewer's suggestion that initial solution quality should be comparable across all variants, and VNS should not be applied prior to the start of the actual evolutionary process. Therefore, to ensure a fair and consistent comparison, we regenerated the experimental results using the same initial population without applying VNS at generation 0 for all three variants.

Based on 20 independent runs, we revised Fig. 19 to present the updated convergence curves generated using the same initial population without applying VNS. In addition, we added a new figure (Fig. 20), which provides a statistically robust boxplot showing the distribution of the final makespan values for each method. These figures together offer a clearer and fairer evaluation of the effectiveness of each algorithm component.

Also, as suggested, the sentence "*The same initial population is used to examine the differences in the search process of each method.*" has been added to Section 4.4 of the revised manuscript.

We hope this revision satisfactorily addresses the reviewer's concern.

Specific modifications in the manuscript:

From revised paper (Section 4.4):

To further validate the effectiveness of the proposed improvements in the IA-NSGA-II, we conducted an ablation study to evaluate the contributions of the adaptive crossover/mutation probability and the VNS. The experiments were conducted using the same test instances and parameter settings as in Experiment 2 to ensure reliability and consistency. **The same initial population is used to examine the differences in the search process of each method.**

From revised paper (Section 4.4):

Fig. 19 shows the convergence curves of the ablation study, while Fig. 20 presents the boxplot comparison of the final makespan for the three variants. The detailed data corresponding to the boxplot is provided in Table 8.

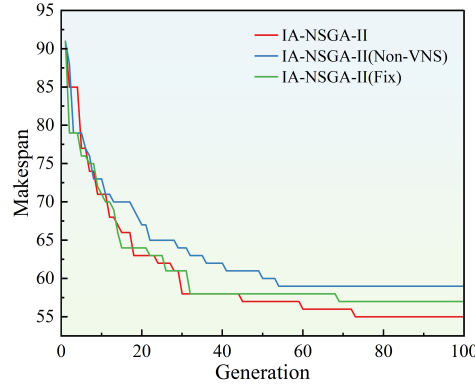


Fig. 19. Convergence curves for IA-NSGA-II variants

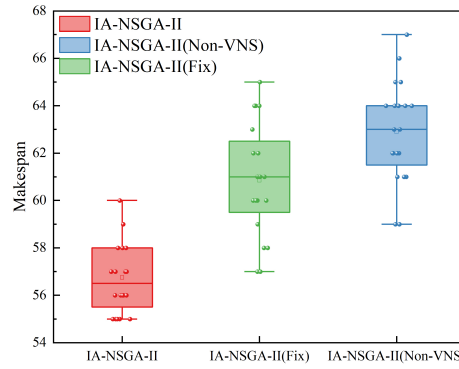


Fig. 20. Boxplot comparison for IA-NSGA-II variants

Table 8. Specific data for ablation study box plot

Box plot structure	<i>IA-NSGA-II</i>	<i>NSGA-II(Fix)</i>	<i>NSGA-II(Non-VNS)</i>
Lower whisker	55	57	58
Lower end of the box	55.5	59.5	61.5
Median	56.5	61	63
Mean	56.75	60.85	62.9
Upper end of the box	58	62.5	64
Upper whisker	61	66	69

The ablation study demonstrates that both the adaptive crossover and mutation probability and the VNS are essential components of IA-NSGA-II. The adaptive probability enhances exploration and exploitation by dynamically adjusting to the population's state, while VNS expands the search space by altering neighborhood structures, helping the algorithm escape local optima. Without the adaptive probability, the algorithm tends to lose diversity in later stages, leading to premature convergence. Without VNS, the search becomes easily trapped in local optima, limiting solution quality. These mechanisms significantly improve performance and collectively contribute to the effectiveness of IA-NSGA-II in solving the FAJSP-AGVs.